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import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import mean_squared_error, r2_score

X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)

y = np.array([2, 5, 10, 17, 26, 37, 50, 65, 82, 101])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

linear_pred = linear_model.predict(X_test)

poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train)

X_test_poly = poly.transform(X_test)

polynomial_model = LinearRegression()

polynomial_model.fit(X_train_poly, y_train)

polynomial_pred = polynomial_model.predict(X_test_poly)

print("Linear Regression")

print("-----")

print("MSE:", mean_squared_error(y_test, linear_pred))

print("R2 Score:", r2_score(y_test, linear_pred))

print("\nPolynomial Regression")

print("-----")

print("MSE:", mean_squared_error(y_test, polynomial_pred))

print("R2 Score:", r2_score(y_test, polynomial_pred))

X_curve = np.linspace(1, 10, 100).reshape(-1, 1)

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linear_curve = linear_model.predict(X_curve)
polynomial_curve = polynomial_model.predict(
    poly.transform(X_curve)
)

plt.scatter(X, y, label="Actual Data")
plt.plot(X_curve, linear_curve, label="Linear Regression")
plt.plot(X_curve, polynomial_curve, label="Polynomial Regression")
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.show()

```

... Linear Regression

MSE: 25.0

R2 Score: 0.9831337493675156

Polynomial Regression

MSE: 1.8972104770565334e-28

R2 Score: 1.0

